

STAT0011: Decision and Risk In-Course Assessment (ICA) 2026

Group 33

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Individual Contributions

23176020

- Constructed the AIC comparison tables for ARMA(p,q)-GARCH(1,1) models across orders 0–5 for both indices to systematically determine the optimal ARMA specification.
- Fitted and compared GARCH, TGARCH, and eGARCH volatility models for both indices using AIC and BIC, selecting TGARCH for its ability to capture the leverage effect.
- Evaluated six candidate conditional distributions for the ARMA-TGARCH models, confirming the skewed Student-t as the best fit for both indices.
- Produced and interpreted diagnostic plots (density, Q-Q, ACF of residuals, ACF of squared residuals) to assess model adequacy.
- Conducted Ljung-Box and Engle’s ARCH tests on the standardised residuals to formally confirm that no remaining autocorrelation or ARCH effects were present.

23014125

- Applied the Probability Integral Transform to the standardised residuals and checked uniformity using histograms and K-S / A-D tests.
- Fitted the copula model and selected the best copula using the AIC criterion.
- Simulated joint observations from the fitted copula using Monte Carlo simulation.
- Transformed simulated uniform values back to skewed Student-t residuals and prepared them to be reintroduced into the ARMA–TGARCH framework for VaR estimation.

24096152

- Conducted preliminary diagnostic checks including ACF/PACF plots, ACF of squared returns, and Engle’s ARCH LM test to confirm the presence of autocorrelation and ARCH effects in both indices, motivating the ARMA-TGARCH framework.
- Derived the conditional mean and variance equations for the S&P500 (ARMA(0,0)-TGARCH(1,1)) and Nikkei 225 (ARMA(3,0)-TGARCH(1,1)) and implemented the one-week ahead simulation by combining fitted model parameters with the copula-generated residuals to produce 10,000 simulated returns per index.
- Validated model fit through long-run simulation plots comparing actual vs. simulated log-return paths for both indices.

24076973

- Selected the indices and provided justification for them.
- Collected the returns, cleaning the dataset to account for discrepancies in holidays.
- Conducted the Exploratory Data Analysis, including the Ljung-Box test.

22019340

- Calculated CVaR and VaR, alongside relevant plots and tables
- Reached out to group and organised meetings
- Assigned tasks to group members and oversaw completion of every section

22016121

- Contributed to the group discussions.
- Wrote the conclusion and contribution list.
- Compiled the final draft, insuring all contributions were consistent. Made grammatical and spelling edits where necessary.

1. Introduction

This report aims to quantify potential downside exposure of an equally weighted portfolio, made up of the S&P 500 and the Nikkei 225, using Value-At-Risk (VaR) metrics at both the 95% and the 99% confidence levels. Weekly returns were analysed between the period of 2000 and 2025.

Our approach uses Monte Carlo simulations, underpinned by Copula theory. We begin with an exploratory data analysis to understand the underlying traits of each indices' log returns. Next we determine the best time series model for each index via methods of comparison. Finally, we fitted and simulated from a chosen Copula to produce an estimate for the portfolio's VaR.

2. Data Selection, Collection & Exploratory Data Analysis

2.1 Data Selection

Our equally-weighted portfolio was constructed using the S&P 500 (USA) and the Nikkei 225 (Japan) indices. The S&P500, made up of the 500 largest US-listed companies, serves as the primary benchmark for the US equity market and is an adequate indicator of global economic health. The Nikkei 225, listed on the Tokyo Stock Exchange, was selected to provide both geographical and macroeconomic contrast. Japan's distinctive macroeconomic conditions set it apart from the US, characterised by decades of near-zero interest rates and chronic deflation (Nishizaki et al, 2012). Leveraging their differences in responding to global shocks, the lower positive correlation between the US and Japanese markets increases our portfolio's diversification benefit.

2.2 Data Collection

Daily closing prices were downloaded from Yahoo Finance between the period 2000-01-01 and 2025-12-31, which were subsequently converted into weekly prices. Logarithmic returns were used, calculated by $r_t = \ln(P_t/P_{t-1})$, because of their time-additivity and to improve the stationarity of the time series.

To account for the disparities in time zones and holidays between the New York and Tokyo exchanges, any weeks containing missing data for either index were removed.

```
# Defining tickers
s_ticker <- "^GSPC"      # S&P 500
n_ticker <- "^N225"      # Nikkei 225

# Downloading data from Yahoo Finance
getSymbols(c(s_ticker, n_ticker), src = "yahoo", from = "2000-01-01", to = "2025-12-31")

# Obtaining weekly prices
ep1 <- endpoints(GSPC, on="weeks")
ep2 <- endpoints(N225, on="weeks")

s_weekly_prices <- Ad(GSPC)[ep1, ]
n_weekly_prices <- Ad(N225)[ep2, ]

# Compute Log Returns
s_log_ret <- diff(log(s_weekly_prices))[-1]
n_log_ret <- diff(log(n_weekly_prices))[-1]

# Standardising dates
```

```

index(s_log_ret) <- as.Date(index(s_log_ret))
index(n_log_ret) <- as.Date(index(n_log_ret))

# Merge indices and remove non-overlapping trading periods between the US and Japan
combined_data <- merge(s_log_ret, n_log_ret, all = FALSE)

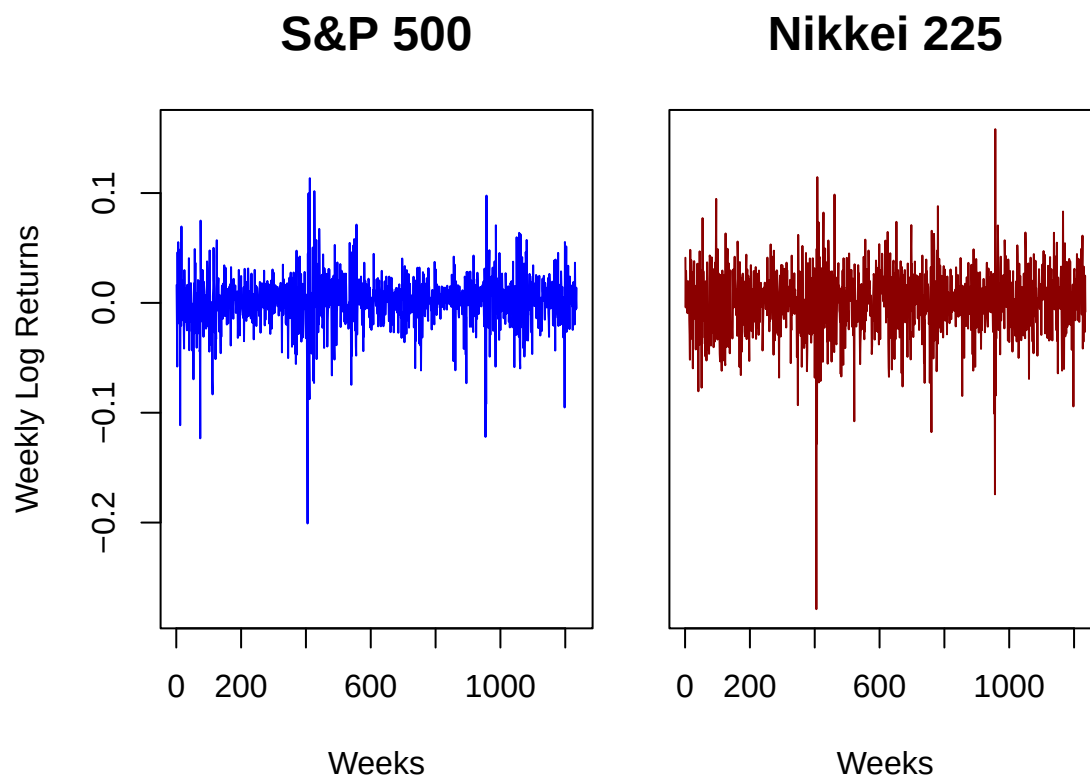
# Omitting any 'NA' values
combined_data <- na.omit(combined_data)
colnames(combined_data) <- c("SP500_Returns", "Nikkei225_Returns")

# Construct Portfolio and Convert to Dataframe for Excel
final_df <- data.frame(Date = index(combined_data), coredata(combined_data))
final_df$Weighted_Portfolio <- 0.5 * (final_df$SP500_Returns + final_df$Nikkei225_Returns)

```

2.3 Exploratory Data Analysis

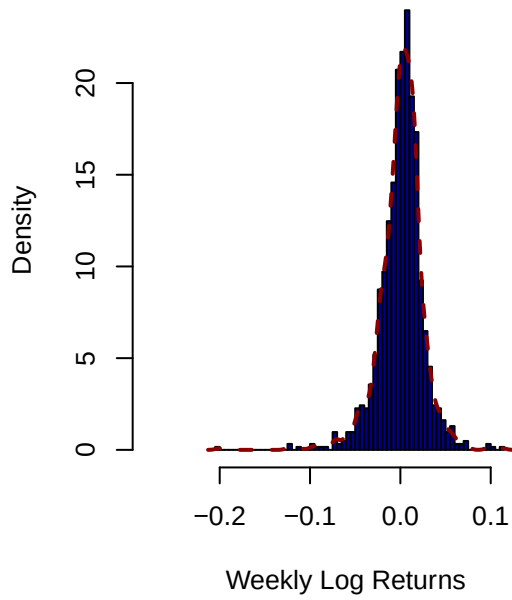
Our dataset contains 1235 weekly log returns from 2000 to 2025. To begin, we visualise the log returns.



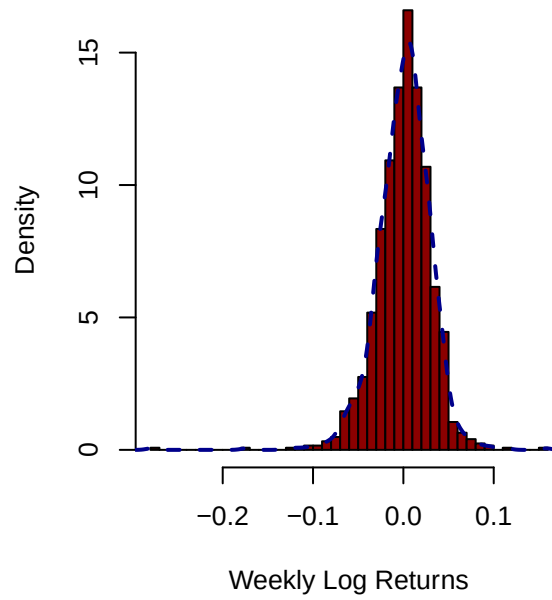
The time series plots reveal that the Nikkei 225 exhibits higher overall volatility and more extreme negative tail events than the S&P 500. Both indices show clear signs of volatility clustering, where large returns (of either sign) are followed by further large returns. The occurrence of simultaneous major downward spikes from both indices suggests a degree of lower tail dependence, which will be taken into account when selecting an appropriate Copula.

We then plotted histograms and Q-Q plots to assess the distribution of the weekly log returns and attempt to discover any deviations from normality.

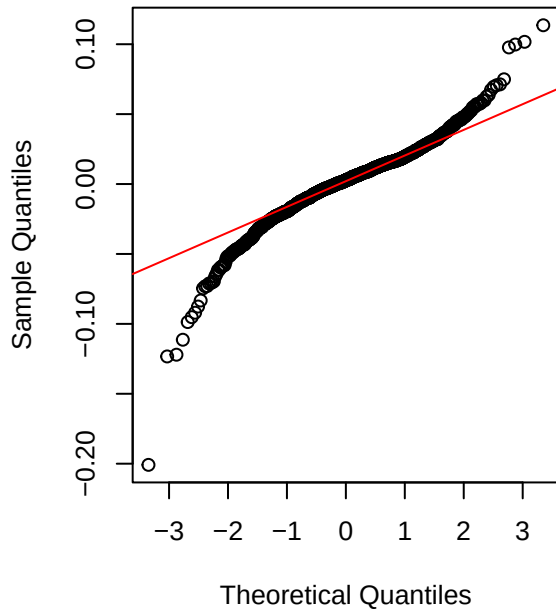
S&P 500 Histogram



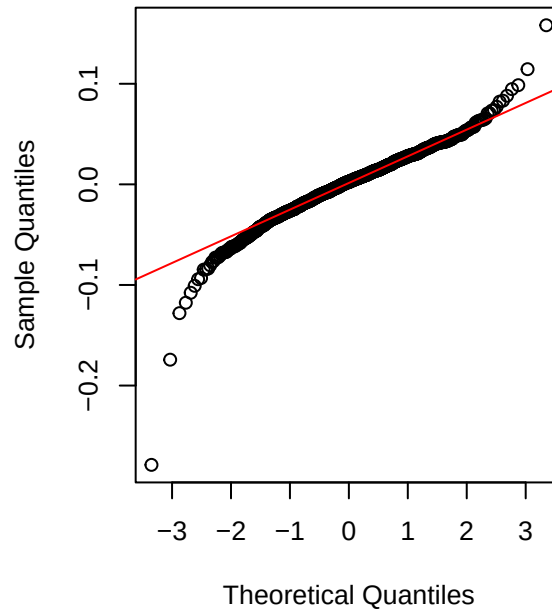
Nikkei 225 Histogram



S&P 500 Q-Q Plot



Nikkei 225 Q-Q Plot



While both indices are centred around zero, they exhibit leptokurtosis, with a sharper peak and fatter tails than a normal distribution. The Q-Q plots confirm that the tails of both indices are significantly thicker than that of a normal distribution, shown by the S-shaped deviation from the red reference line. This deviation is particularly pronounced in the lower left quadrant, indicating that the risk of a crash (extreme negative returns) is much higher than a normal distribution would suggest.

Collectively, the plots demonstrate that the weekly log returns are non-normally distributed. This further necessitates the use of a model that can accurately account for the joint risk and tail dependence between the two indices.

To assess whether the historical returns of our chosen indices possess predictive power over future performance, we conduct the Ljung-Box test on the log-return series. We examine three distinct time horizons: 13 weeks lag (quarterly), 26 weeks lag (semi-yearly), and 52 weeks lag (annual). These intervals allow us to detect potential cyclical dependencies or seasonal patterns within the data.

The null hypothesis (H_0) for the Ljung-Box test is that the returns are independently and identically distributed, indicating an absence of serial correlation up to the specified lag. A resulting p-value below our significance threshold of 0.05 would lead us to reject H_0 , suggesting that the returns are not purely random and that an ARMA framework may be required to model the conditional mean.

Table 1: Ljung-Box Test Results for Log-returns

Metric	SP500	Nikkei225
Quarterly	29.8958537	16.3778551
p-value	0.0048752	0.2293264
Half-yearly	62.6494366	26.1507946
p-value	0.0000731	0.4548401
Anually	96.0683782	51.6235320
p-value	0.0001945	0.4886454

As summarized above, the S&P 500 exhibits highly significant autocorrelation at every lag, with p-values ranging from 0.0048 to 0.00019, leading us to reject the null hypothesis of independence and suggesting a persistent return structure that necessitates an ARMA model. In contrast, the Nikkei 225 results demonstrate no statistically significant serial dependency, with p-values consistently exceeding the 0.05 threshold (from 0.229 to 0.488), indicating that the Japanese index follows a stochastic process more aligned with a random walk. Collectively, these findings confirm that while the Nikkei's mean is independent, the significant departures from randomness in the S&P 500, combined with the observed non-normality in both series, require a more sophisticated modeling approach, such as GARCH, to capture the volatility clustering and joint tail dependence essential for accurate Value-at-Risk estimation.

3. Marginal Distribution Modelling

3.1 Fitting an ARMA Model

3.1.1 AIC Table Analysis

To systematically determine the optimal ARMA order for each index, we computed AIC values for all ARMA(p,q)-GARCH(1,1) combinations with $p, q \in \{0, 1, \dots, 5\}$. Although the GARCH(1,1) component is introduced prematurely at this stage, it is included to ensure that the AIC comparison is not distorted by unmodelled volatility clustering (which, as established in the EDA, is present in both series). The conditional distribution is set to the skewed Student-t (**sstd**) throughout, for reasons that will be confirmed later.

Table 2: SP500 ARMA(p,q)-GARCH(1,1) AIC Values

	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)	MA(5)
AR(0)	-4.9232	-4.9372	-4.9417	-4.9440	-4.9450	-4.9440
AR(1)	-4.9348	-4.9458	-4.9440	-4.9488	-4.9493	-4.9454

	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)	MA(5)
AR(2)	-4.9377	-4.9441	-4.9437	-4.9484	-4.9477	-4.9416
AR(3)	-4.9443	-4.9458	-4.9476	-4.9469	-4.9501	-4.9499
AR(4)	-4.9441	-4.9466	-4.9472	-4.9484	-4.9486	-4.9483
AR(5)	-4.9443	-4.9446	-4.9458	-4.9474	-4.9473	-4.9517

S&P500 AIC Table

The AIC values for the S&P500 are tightly clustered across all ARMA orders. The lowest AIC is achieved at ARMA(5,3), but the improvement over ARMA(0,0) is marginal (approximately 0.037). Beyond ARMA(0,0), increasing the ARMA order yields negligible gains, with the values forming a near-horizontal surface. This suggests that there is very little linear dependence in the S&P500 weekly log-returns to be captured by an ARMA component. Consequently, ARMA(0,0) is a strong candidate as it provides comparable effectiveness whilst maintaining maximum parsimony.

Table 3: N225 ARMA(p,q)-GARCH(1,1) AIC Values

	MA(0)	MA(1)	MA(2)	MA(3)	MA(4)	MA(5)
AR(0)	-4.3175	-4.3188	-4.3200	-4.3230	-4.3226	-4.3218
AR(1)	-4.3188	-4.3173	-4.3175	-4.3218	-4.3224	-4.3214
AR(2)	-4.3196	-4.3175	-4.3184	-4.3202	-4.3213	-4.3198
AR(3)	-4.3230	-4.3214	-4.3199	-4.3199	-4.3198	-4.3218
AR(4)	-4.3225	-4.3224	-4.3214	-4.3198	-4.3182	-4.3204
AR(5)	-4.3219	-4.3213	-4.3198	-4.3216	-4.3202	-4.3217

Nikkei 225 AIC Table

A similar pattern emerges for the Nikkei 225: the AIC values are very tightly clustered, with the total range spanning approximately 0.009. The lowest AIC is achieved at ARMA(3,0), and the AR(3) row generally performs well across different MA orders. However, adding MA terms beyond MA(0) provides no meaningful improvement, indicating that the MA component is unnecessary. The slight advantage of ARMA(3,0) over ARMA(0,0) suggests a possible autoregressive effect at lag 3, which we will investigate further using the ACF/PACF plots.

3.1.2 ARMA Results

From the analysis on the weekly log-returns of the S&P500, we have chosen an **ARMA(0,0)** model because:

- The ACF and PACF plots show no significant spikes beyond lag 0, indicating no meaningful linear dependence in the returns. This is consistent with the efficient market hypothesis, where returns on a major, highly liquid index like the S&P500 are difficult to predict from past values.
- The AIC table confirms that increasing the ARMA order beyond (0,0) yields negligible improvement.
- It is the most parsimonious specification, and there is insufficient evidence to justify additional parameters.

From the analysis on the weekly log-returns of the Nikkei 225, we have chosen an **ARMA(3,0)** model because:

- The PACF displays a notable spike at lag 3, suggesting an autoregressive effect at that lag.
- The AIC table confirms that ARMA(3,0) achieves the lowest AIC value across the entire grid, and that adding MA terms provides no further improvement.
- It provides a good balance between parsimony and explanatory power: the AR(3) component captures the lag-3 dependence without over-parameterising the model with unnecessary MA terms.

3.2 Fitting a GARCH Type Model

We previously found evidence of volatility clustering and significant ARCH effects in our log-returns. This led us to the conclusion that a GARCH-type model is necessary to capture the time-varying volatility.

3.2.1 Model Parameters (ARCH and GARCH Terms)

From the lecture notes: “In practice, when people (both in academia and industry – banks, hedge funds, etc) use a GARCH model for time-varying conditional variance, it is overwhelmingly the GARCH(1,1) model which is chosen.” The ACF plots of squared log-returns (provided in the EDA section) further support this: both indices show strong immediate impact of volatility shocks (large spike at lag 1) followed by gradual decay, indicating that a single ARCH and a single GARCH parameter are sufficient to capture the immediate and persistent effects of volatility shocks. We will later analyse the squared residuals of our model, and if necessary (i.e., if the squared residuals still display autocorrelation), revise our parameter estimates.

3.2.2 GARCH, TGARCH, eGARCH Model Comparison

In this section, we compare three variants of volatility models: standard GARCH, TGARCH, and eGARCH. While the standard GARCH model captures volatility clustering through symmetric responses to shocks, both TGARCH and eGARCH allow for asymmetry in the volatility process. The TGARCH model explicitly models the leverage effect by allowing negative shocks to have a larger impact on volatility than positive shocks. This is generally observable in financial assets and supported by the Volatility Feedback Hypothesis, which suggests that falling prices spur more selling and thus increase volatility, while rising prices boost investor confidence and reduce uncertainty. Similarly, the eGARCH model captures asymmetry through a logarithmic formulation, ensuring that the conditional variance is always positive. Given that the log-returns of the S&P500 and Nikkei 225 exhibit heavier negative tails and asymmetric responses, particularly during market downturns, these models might better accommodate the observed asymmetries.

We fitted three models for each index with the same specifications (ARMA order, GARCH(1,1) parameters, and skewed Student-t conditional distribution), varying only the type of volatility model. We base our selection on AIC and BIC, looking for the best balance between fit and parsimony.

Comparing the models using AIC and BIC reveals that the standard GARCH specification is outperformed by both the TGARCH and eGARCH models in capturing the asymmetric volatility features of the data. For the S&P500, the ARMA-TGARCH model exhibits the lowest AIC and BIC values, indicating a better fit compared to both the standard GARCH and the eGARCH alternatives. For the Nikkei 225, the differences between TGARCH and eGARCH are marginal. Given that the TGARCH model directly captures the leverage effect – by allowing negative shocks to have a more pronounced impact on volatility – and offers clear interpretability, we opt for the TGARCH specification for consistency across both indices, despite any slight numerical edge for eGARCH.

3.2.3 Conditional Distribution Check

We will now verify whether our choice of the skewed Student-t (`sstd`) conditional distribution is appropriate. To do so, we evaluated the respective ARMA-TGARCH models for each index across six candidate conditional distributions, holding all other model parameters constant (ARMA order, GARCH order). The function iterates through each distribution and returns the corresponding AIC and BIC values.

This comparison is motivated by our earlier diagnostics: the Jarque-Bera test indicated significant departures from normality, and the histograms and Q-Q plots revealed heavy tails and skewness in the log-returns. A standard Normal distribution would therefore be inadequate. The skewed Student-t distribution (`sstd`) is expected to perform well because it accommodates both heavy tails (via the degrees of freedom parameter) and asymmetry (via the skew parameter).


```

cond_dists <- c("norm", "std", "ged", "snorm", "sged", "sstd")

# S&P500
ics_table1 <- matrix(NA, nrow = 6, ncol = 2)
rownames(ics_table1) <- cond_dists
colnames(ics_table1) <- c("AIC", "BIC")

for (i in seq_along(cond_dists)) {
  spec <- ugarchspec(
    variance.model = list(model = "fGARCH", garchOrder = c(1,1), submodel = "TGARCH"),
    mean.model = list(armaOrder = c(0,0), include.mean = TRUE),
    distribution.model = cond_dists[i])
  model <- ugarchfit(data = final_df$SP500>Returns, spec = spec, solver = "solnp")
  ics_table1[i, "AIC"] <- infocriteria(model)[1]
  ics_table1[i, "BIC"] <- infocriteria(model)[2]
}

kable(ics_table1, caption = "Results for S&P500 - ARMA(0,0)-TGARCH(1,1)")

```

Table 4: Results for S&P500 - ARMA(0,0)-TGARCH(1,1)

	AIC	BIC
norm	-4.910257	-4.889533
std	-4.944697	-4.919828
ged	-4.934813	-4.909944
snorm	-4.968958	-4.944090
sged	-4.982429	-4.953415
sstd	-4.985937	-4.956923

```

# N225
ics_table2 <- matrix(NA, nrow = 6, ncol = 2)
rownames(ics_table2) <- cond_dists
colnames(ics_table2) <- c("AIC", "BIC")

for (i in seq_along(cond_dists)) {
  spec <- ugarchspec(
    variance.model = list(model = "fGARCH", garchOrder = c(1,1), submodel = "TGARCH"),
    mean.model = list(armaOrder = c(3,0), include.mean = TRUE),
    distribution.model = cond_dists[i])
  model <- ugarchfit(data = final_df$Nikkei225>Returns, spec = spec, solver = "solnp")
  ics_table2[i, "AIC"] <- infocriteria(model)[1]
  ics_table2[i, "BIC"] <- infocriteria(model)[2]
}

kable(ics_table2, caption = "Results for N225 - ARMA(3,0)-TGARCH(1,1)")

```

Table 5: Results for N225 - ARMA(3,0)-TGARCH(1,1)

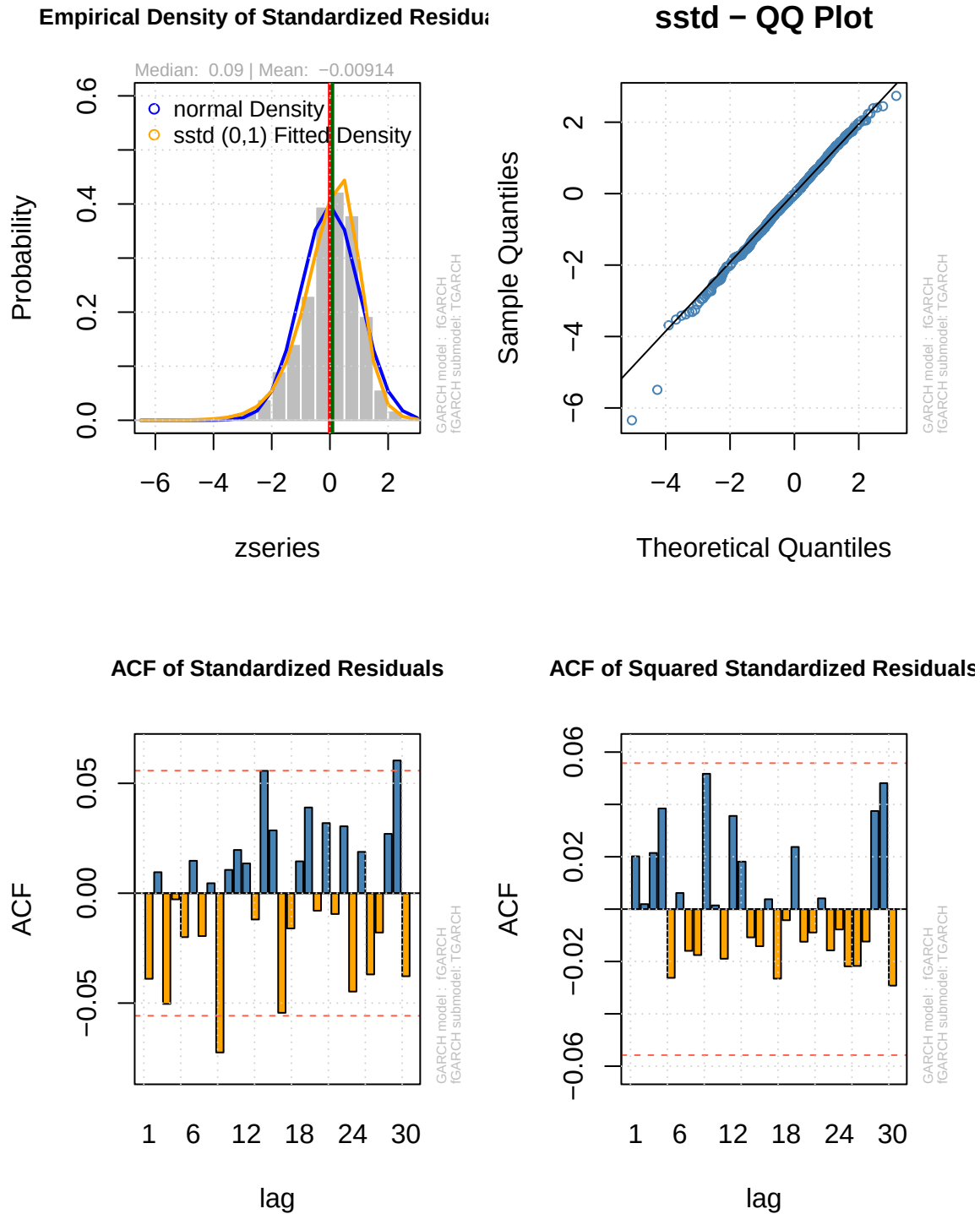
	AIC	BIC
norm	-4.302887	-4.269729

	AIC	BIC
std	-4.322540	-4.285236
ged	-4.315375	-4.278072
snorm	-4.324810	-4.287507
sged	-4.334595	-4.293147
sstd	-4.338870	-4.297421

For both indices, the skewed Student-t distribution (**sstd**) achieves the lowest AIC and BIC values, confirming our earlier intuition. The ranking tells a coherent story: skewed distributions (**snorm**, **sged**, **sstd**) consistently outperform their non-skewed counterparts (**norm**, **ged**, **std**), confirming that skewness is an important feature of the data. Similarly, heavy-tailed distributions (**std**, **sstd**) outperform their lighter-tailed equivalents (**norm**, **snorm**), confirming the relevance of excess kurtosis. The **sstd** distribution captures both features simultaneously, which is why it provides the best fit.

3.3 Model Checking

3.3.1 S&P500 Diagnostic Plots



Distributional Fit Diagnostics

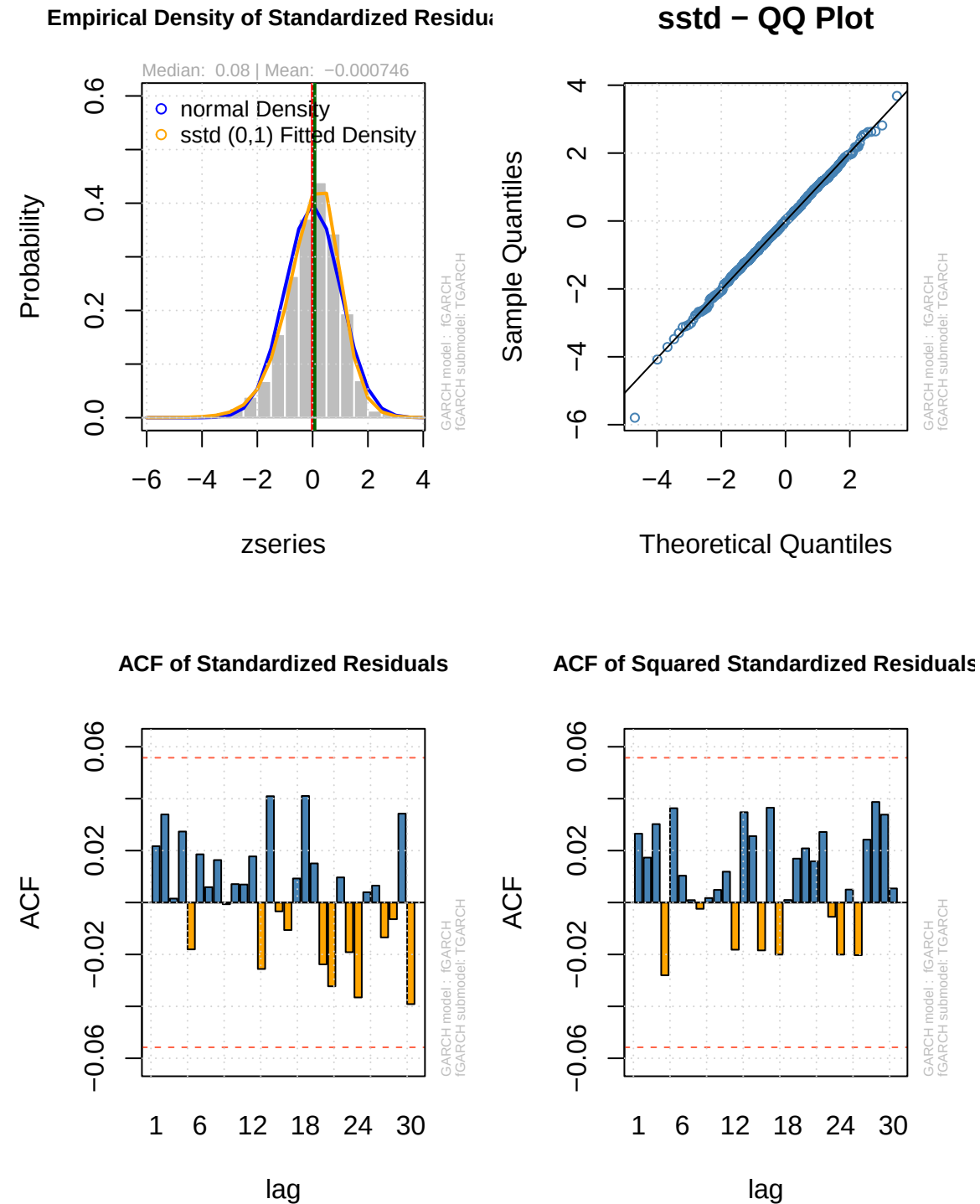
The first plot shows the empirical density (histogram) of the standardised residuals, overlaid with the theoretical density curves of the Normal and **sstd** distributions. As previously concluded, the **sstd** distribution provides a better fit: the histogram reveals that the residuals exhibit heavier negative tails (i.e., asymmetry), and the **sstd** density curve fits more closely, indicating that the skew parameter is helping to capture that asymmetry.

The Q-Q plot compares the quantiles of our standardised residuals to those of the fitted **sstd** distribution. Almost all points are tightly clustered along the 45-degree line, with no systematic deviation patterns, indicating a very good fit. Any outliers in the extreme lower tail could reflect rare extreme events (such as the 2008 financial crisis or the Covid market crash), which are plausible given the 25-year span of our data.

Autocorrelation and ARCH Effects Diagnostics

The ACF plots of the standardised residuals and squared standardised residuals are used to assess whether the model has successfully captured the linear dependence in the mean and the volatility clustering respectively. In both plots, nearly all bars lie within the dashed significance bounds, indicating that neither the standardised residuals nor the squared residuals display significant autocorrelation. We can thus conclude that the ARMA(0,0)-TGARCH(1,1) model has successfully captured both the linear autocorrelation (or lack thereof) in the returns and the ARCH effects present in our data.

3.3.2 Nikkei 225 Diagnostic Plots



Distributional Fit Diagnostics

Again, the histogram of standardised residuals aligns more closely with the skewed Student-t density curve than with the Normal one, indicating that the skew parameter effectively accounts for the asymmetry

in the tails of the returns distribution, while also accommodating heavier tails than those of the Normal distribution. The Q-Q plot shows that most points lie near the 45-degree line, confirming that the majority of the standardised residuals conform well with the assumed skewed Student-t distribution. Although a small number of points in the extreme tails deviate from the theoretical line, the overall alignment is strong. Thus, we conclude that the ARMA(3,0)-TGARCH(1,1) model, with a skewed Student-t specification, adequately captures the key features of the Nikkei 225 return distribution, including its skewness and heavier tails.

Autocorrelation and ARCH Effects Diagnostics

Looking at the ACF of the standardised residuals, most bars lie within the significance bounds, and there is no strong evidence of remaining autocorrelation. Any isolated spikes at high lags are likely attributable to chance (with 30 lags, we expect approximately 1-2 spurious significant spikes at the 5% level). In the ACF of squared standardised residuals, nearly all bars are within the bounds. This indicates that the TGARCH component has adequately captured the volatility clustering. To formally confirm these visual findings, we conduct the Ljung-Box and ARCH tests below.

3.3.3 Ljung-Box and Engle's ARCH Tests

To confirm the results from the diagnostic plots, we conducted Ljung-Box tests on both the standardised residuals and the squared standardised residuals, as well as Engle's ARCH tests for both the S&P500 and Nikkei 225 models. The Ljung-Box test assesses whether there is remaining autocorrelation in the residuals (null hypothesis: no autocorrelation), while Engle's ARCH test evaluates whether there are remaining ARCH effects (null hypothesis: no ARCH effects, i.e., constant variance).

Table 6: Ljung-Box Test Results for Residuals and Squared Residuals

Test	S.P.500	N.225
Statistic Residuals	13.9895066	5.4871921
p-value	0.3745834	0.9628465
Statistic Squared Residuals	10.2971212	7.2646578
p-value	0.6694759	0.8880023

Table 7: Engle's ARCH Test Results for Residuals

Test	S.P.500	N.225
ARCH Test Statistic	7.3097365	7.2872013
p-value	0.8854919	0.8867505

The results show p-values well above the 0.05 significance threshold for both indices across all tests. This means we fail to reject the null hypotheses of no autocorrelation (Ljung-Box) and no ARCH effects (Engle's ARCH test) in the standardised residuals. This confirms that there is no remaining autocorrelation or ARCH effect in the residuals of either model, indicating that both the ARMA(0,0)-TGARCH(1,1) model for the S&P500 and the ARMA(3,0)-TGARCH(1,1) model for the Nikkei 225 are well specified.

4. Copula Modelling

4.1 Probability Integral Transformation

```

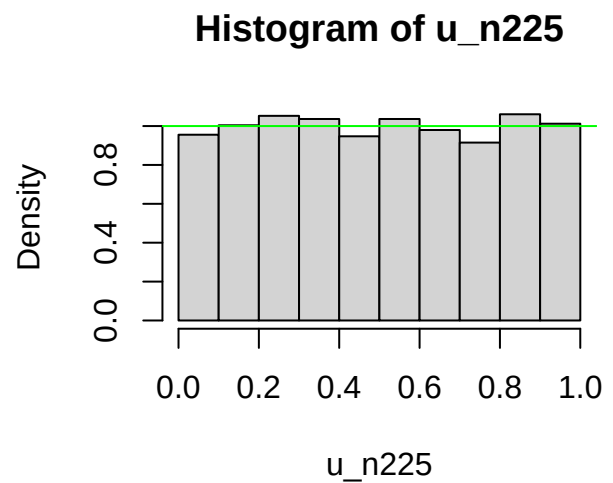
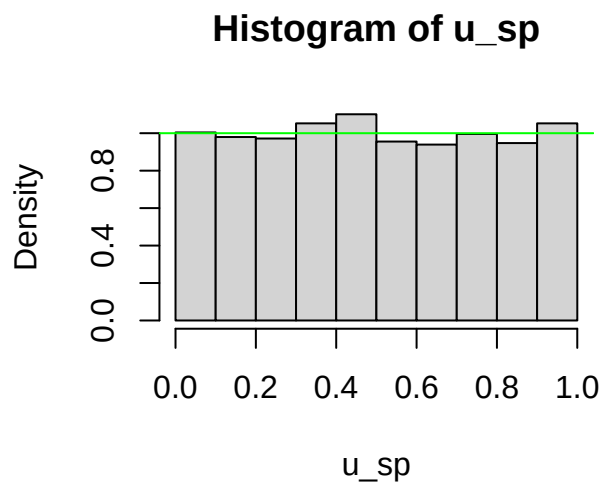
# Extract skewed Student-t parameters from the fitted TGARCH models

# S&P 500 parameters
sp_shape <- coef(SP500_model2)["shape"]
sp_skew  <- coef(SP500_model2)["skew"]

# Nikkei 225 parameters
n225_shape <- coef(N225_model2)["shape"]
n225_skew  <- coef(N225_model2)["skew"]

# Apply PIT to standardised residuals
u_sp <- psstd(as.numeric(SP500_res), nu = sp_shape, xi = sp_skew)
u_n225 <- psstd(as.numeric(N225_res), nu = n225_shape, xi = n225_skew)

```



```

# K-S tests for Uniform(0,1)

KS_sp <- LcKS(as.numeric(u_sp), cdf = "punif")
KS_n225 <- LcKS(as.numeric(u_n225), cdf = "punif")

# A-D tests for Uniform(0,1)

AD_sp <- ad.test(as.numeric(u_sp), null="punif")
AD_n225 <- ad.test(as.numeric(u_n225), null="punif")

# Combine p-values into a table for reporting

uniform_test_results <- data.frame(
  Test = c("K-S p-value", "A-D p-value"),
  `S&P 500` = c(unname(KS_sp$p.value), unname(AD_sp$p.value)),
  `Nikkei 225` = c(unname(KS_n225$p.value), unname(AD_n225$p.value)),
  check.names = FALSE
)

```

```
rownames(uniform_test_results) <- NULL

kable(
  uniform_test_results,
  caption = "Kolmogorov-Smirnov and Anderson-Darling Tests for Uniformity",
  row.names = FALSE
)
```

Table 8: Kolmogorov-Smirnov and Anderson-Darling Tests for Uniformity

Test	S&P 500	Nikkei 225
K-S p-value	0.8330000	0.9752000
A-D p-value	0.9771833	0.9969413

Copulas are defined on the unit square $[0,1]^2$, meaning each marginal distribution must first be transformed to a Uniform(0,1) distribution. To achieve this, we apply the Probability Integral Transformation (PIT) to the standardised residuals obtained from each fitted model.

The histograms of the transformed residuals for both the S&P500 and the Nikkei 225 appear approximately flat over the interval $[0,1]$, with no clear concentration near 0 or 1. This is consistent with the behaviour expected under a Uniform(0,1) distribution and suggests that the marginal models provide an adequate fit. Applying the skewed Student-t CDF to the standardised residuals produces variables that behave like Uniform(0,1) observations. As a result, the skewed Student-t assumption for the residuals appears to be adequate in capturing the marginal distributional features of the data, including skewness and heavy tails.

Furthermore, we performed Kolmogorov-Smirnov and Anderson-Darling tests for uniformity. For both indices, the p-values from the K-S and A-D tests are well above the 5% significance level. Therefore, we fail to reject the null hypothesis that the transformed residuals follow a Uniform(0,1) distribution. This means the PIT-transformed residuals can be treated as Uniform(0,1) variables, allowing us to proceed to modelling the dependence structure using copulas. Since uniformity of the PIT-transformed residuals relies on correct specification of the marginal distributions, these results are consistent with the skewed Student-t assumption used in the fitted marginal models.

4.2 Copula Fitting & Monte Carlo Simulation

```
# Make both PIT series have same length

n <- min(length(u_sp), length(u_n225))
u_sp <- u_sp[seq_len(n)]
u_n225 <- u_n225[seq_len(n)]

# Combine into a data frame and remove any remaining missing values

copula_data <- na.omit(data.frame(u_sp = as.numeric(u_sp),
                                  u_n225 = as.numeric(u_n225)))

# Avoid exact 0 or 1 values (these can cause numerical issues)

eps <- 1e-10
```



```

copula_data$u_sp <- pmin(pmax(copula_data$u_sp, eps), 1 - eps)
copula_data$u_n225 <- pmin(pmax(copula_data$u_n225, eps), 1 - eps)

# Select the best copula using AIC

copula_fit <- VineCopula::BiCopSelect(
  copula_data$u_sp,
  copula_data$u_n225,
  familyset = c(0, 1, 2, 3, 4, 5, 6, 7),
  selectioncrit = "AIC",
  indeptest = TRUE,
  level = 0.05,
  se = TRUE,
  rotations = FALSE
)

copula_fit

## Bivariate copula: t (par = 0.55, par2 = 17.39, tau = 0.37)

# Value-at-Risk using Monte Carlo simulation
N <- 10000
u_sim <- BiCopSim(N, family=copula_fit$family, copula_fit$par, copula_fit$par2)

# Transform simulated uniforms back to skewed Student-t innovations
res_sp_sim <- qsst(u_sim[,1], nu=sp_shape, xi = sp_skew)
res_n225_sim <- qsst(u_sim[,2], nu=n225_shape, xi = n225_skew)

```

To select an appropriate copula, we use the `BiCopSelect` function from the `VineCopula` package. This function automatically compares a range of candidate bivariate copula families and selects the most suitable model according to a specified information criterion. In our case, the selection criterion is the AIC, meaning that the copula with the lowest AIC is chosen. Additionally, the argument `indeptest = TRUE` ensures that an initial independence test is performed. If independence is rejected at the 5% level, the function will estimate and compare alternative copula models. Based on this data-driven selection procedure, the copula chosen is the Student-t copula.

The Student-t copula is suitable because it allows for symmetric tail dependence, meaning that extreme gains or losses in returns can occur more frequently than under a Gaussian copula. The estimated correlation parameter indicates a moderate positive dependence between the two indices. The degrees of freedom parameter controls the heaviness of the joint tails; the estimate suggests that extreme movements in the two indices may occur together more often than under a Gaussian copula. The implied Kendall's tau supports the presence of a moderate level of association between the two return series.

To estimate portfolio risk measures, we simulate $n = 10,000$ pairs of observations from the fitted Student-t copula using the estimated parameters. The simulation produces a two-column matrix, where each row corresponds to one simulated joint observation. Although each pair of simulated uniform values is generated independently across rows, the dependence within each pair reflects the structure imposed by the fitted copula.

At this stage, the simulated values are still on the $\text{Uniform}(0,1)$ scale. To recover simulated innovations consistent with our marginal models, we transform these uniform draws back using the inverse CDF of the skewed Student-t distribution. This step ensures that the simulated innovations preserve the marginal behaviour captured by the skewed Student-t distribution (skewness and heavy tails) and the cross-sectional dependence captured by the Student-t copula.

These simulated residuals are then ready to be reintroduced into the fitted marginal models to generate simulated one-week ahead returns and compute the portfolio Value-at-Risk.

5. Re-introducing Autocorrelation, Moving Average and GARCH Effects

Before selecting and fitting our ARMA-TGARCH models, we first examine the autocorrelation structure and volatility properties of the raw weekly log-returns for both indices.

5.0 Preliminary Checks

5.0.1 ACF and PACF of Raw Log-Returns

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) indicate whether an ARMA structure is warranted: significant ACF spikes suggest a moving average component, while significant PACF spikes suggest an autoregressive component.

```
par(mfrow = c(2, 2))

acf(final_df$SP500_Returns, lag.max = 30,
    main = "ACF - S&P500 Weekly Log-Returns",
    col = "darkblue", lwd = 2)

pacf(final_df$SP500_Returns, lag.max = 30,
    main = "PACF - S&P500 Weekly Log-Returns",
    col = "darkblue", lwd = 2)

acf(final_df$Nikkei225_Returns, lag.max = 30,
    main = "ACF - Nikkei 225 Weekly Log-Returns",
    col = "darkred", lwd = 2)

pacf(final_df$Nikkei225_Returns, lag.max = 30,
    main = "PACF - Nikkei 225 Weekly Log-Returns",
    col = "darkred", lwd = 2)
```

```
par(mfrow = c(1, 1))
```

For the S&P500, no significant spikes are present in either the ACF or PACF, suggesting no meaningful autocorrelation structure in the returns, consistent with our selection of ARMA(0,0). For the Nikkei 225, significant spikes in the PACF at early lags but not in the ACF suggest a pure autoregressive structure, consistent with our selection of ARMA(3,0).

5.0.2 Evidence of ARCH Effects

Significant autocorrelation in the *squared* returns is a standard indicator of volatility clustering and ARCH effects. We first visualise this, then confirm it formally with Engle's ARCH LM test. We use 13 lags, corresponding to approximately one quarter of weekly data, which is a standard convention for capturing short-to-medium term volatility persistence without overfitting.

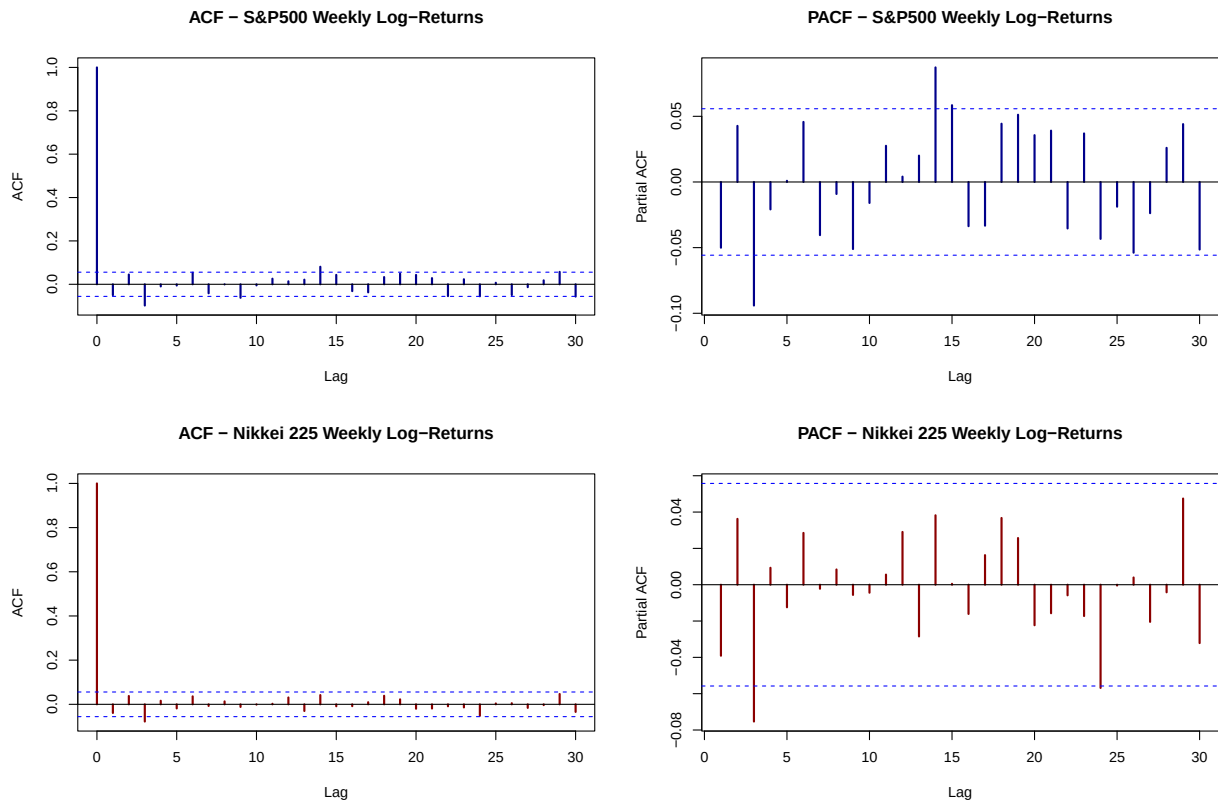


Figure 1: ACF and PACF of raw weekly log-returns for the S&P500 (top row) and Nikkei 225 (bottom row). Blue dashed lines represent 95% confidence bounds under the null hypothesis of no autocorrelation.

```

par(mfrow = c(1, 2))

acf(final_df$SP500>Returns^2, lag.max = 30,
    main = "ACF of Squared Returns - S&P500",
    col = "darkblue", lwd = 2)

acf(final_df$Nikkei225>Returns^2, lag.max = 30,
    main = "ACF of Squared Returns - Nikkei 225",
    col = "darkred", lwd = 2)

```

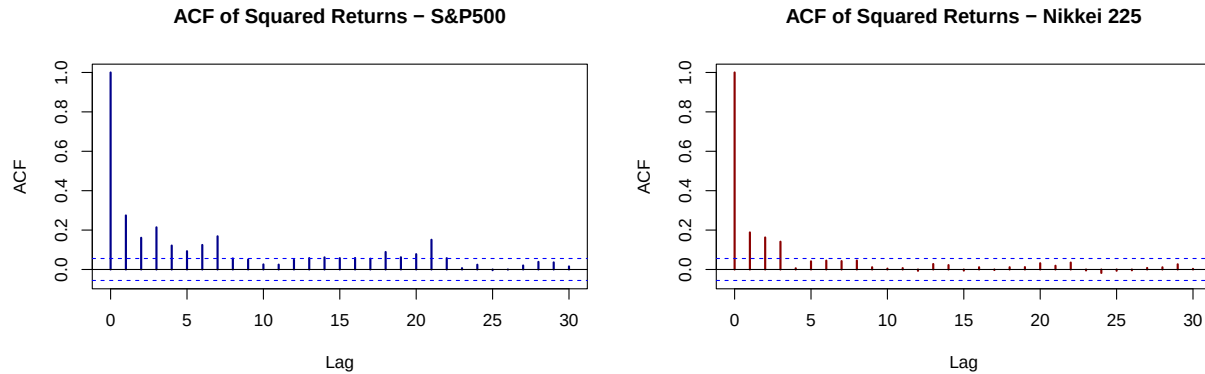


Figure 2: ACF of squared weekly log-returns for the S&P500 (left) and Nikkei 225 (right). Significant spikes across many lags confirm the presence of ARCH effects in both series.

```

par(mfrow = c(1, 1))

# Engle's ARCH LM Test on raw log-returns (13 lags)
arch_test_raw_SP    <- ArchTest(final_df$SP500>Returns, lags = 13)
arch_test_raw_N225  <- ArchTest(final_df$Nikkei225>Returns, lags = 13)

results_arch_raw <- data.frame(
  Test      = c("ARCH LM Statistic", "p-value"),
  `S&P 500` = c(round(arch_test_raw_SP$statistic, 4),
                 format.pval(arch_test_raw_SP$p.value, digits = 4)),
  `Nikkei 225` = c(round(arch_test_raw_N225$statistic, 4),
                    format.pval(arch_test_raw_N225$p.value, digits = 4))
)

kable(results_arch_raw,
      caption = "Engle's ARCH LM Test on Raw Weekly Log-Returns (13 lags)")

```

Table 9: Engle's ARCH LM Test on Raw Weekly Log-Returns (13 lags)

	Test	S.P.500	Nikkei.225
Chi-squared	ARCH LM Statistic	165.6266	81.8857
	p-value	< 2.2e-16	4.868e-12

Both series exhibit highly significant autocorrelation in squared returns and extremely small p-values from the ARCH LM test, providing overwhelming evidence to reject the null hypothesis of no ARCH effects. This confirms that the conditional variance of both return series is time-varying, jointly motivating the use of an ARMA-TGARCH model for both indices.

To re-introduce the appropriate autocorrelation, moving average and GARCH effects for each index, we then followed this process:

- Obtained the 10,000 simulated non-standardised residuals (ϵ) from the copula model.
- Extracted the required parameters from the original ARMA-TGARCH models.
- Simulated weekly returns by:
 1. Calculating the conditional volatility and mean using the ARMA-TGARCH model parameters, the final non-standardised residual values, the sigma values from the respective models, and the last weekly log-return(s) from the data.
 2. Combining the calculated conditional mean and volatility with the simulated residuals from the copula to generate the final simulated one-week return.

5.1 For the S&P500

5.1.1 Explanation

Our final model for the S&P500 was an ARMA(0,0)-TGARCH(1,1) which results in the following formulae for the simulated returns (y_t):

$$y_t = \mu + u_t \quad \text{where} \quad u_t = \sigma_t \epsilon_t, \quad \mu = c$$

In this case, c is assigned the value of the estimated constant term (drift term μ) from the model. There are no autoregressive or moving average terms — the conditional mean is simply a constant, meaning past returns and past errors do not help predict future returns.

$$\sigma_t^2 = \omega + \alpha_1 u_{t-1}^2 + \gamma_1 u_{t-1}^2 I[u_{t-1} < 0] + \beta_1 \sigma_{t-1}^2$$

Here, the conditional variance is related not only to the previous value of the squared error terms, but also to the conditional variance from the previous period as well as the Leverage effect, given by γ which captures the asymmetric impact of negative past errors.

5.1.2 Code

```
# Extract the non-standardised residuals
res1_nonstd <- residuals(SP500_model2, standardize = FALSE)

##### Simulate 1 week ahead for the S&P500 #####

y_last1      <- tail(final_df$SP500>Returns, 1)
r_last1      <- as.numeric(tail(res1_nonstd, 1))
sigma_last1  <- as.numeric(tail(sigma(SP500_model2), 1))

# TGARCH(1,1) conditional volatility
sigma_sim1 <- sqrt(
  coef(SP500_model2)["omega"] +
```

```

coef(SP500_model12)["alpha1"] * r_last1^2 +
coef(SP500_model12)["eta11"] * r_last1^2 * (r_last1 < 0) +
coef(SP500_model12)["beta1"] * sigma_last1^2
)

# ARMA(0,0) conditional mean - constant only
mu_t1 <- coef(SP500_model12)["mu"]

# Simulated return (10,000 scenarios)
y1_1week <- mu_t1 + sigma_sim1 * res_sp_sim

```

5.2 For the Nikkei 225

5.2.1 Explanation

Our final model for the Nikkei 225 was an ARMA(3,0)-TGARCH(1,1) which results in the following formulae for the simulated returns (y_t):

$$y_t = \mu + u_t \quad \text{where} \quad u_t = \sigma_t \epsilon_t, \quad \mu = c + \phi_{(1)} y_{t-1} + \phi_{(2)} y_{t-2} + \phi_{(3)} y_{t-3}$$

In this case, c is also assigned the value of the estimated constant term (drift term μ) from the model. There are no moving average terms; instead $\phi^{(i)}$ is the i^{th} $ar(i)$ term where $i \in \{1, 2, 3\}$, capturing dependence on past return values only.

$$\sigma_t^2 = \omega + \alpha_1 u_{t-1}^2 + \gamma_1 u_{t-1}^2 I[u_{t-1} < 0] + \beta_1 \sigma_{t-1}^2$$

The same theoretical framework applies here, as the TGARCH models for both indices share the same structure.

5.2.2 Code

```

# Extract the non-standardised residuals
res2_nonstd <- residuals(N225_model12, standardize = FALSE)

##### Simulate 1 week ahead for the Nikkei 225 #####

# Extract the three most recent return values for the AR(3) terms
n2 <- nrow(final_df)
y_last2 <- final_df$Nikkei225>Returns[n2] # y_{t-1}
y_last2_lag2 <- final_df$Nikkei225>Returns[n2 - 1] # y_{t-2}
y_last2_lag3 <- final_df$Nikkei225>Returns[n2 - 2] # y_{t-3}

r_last2 <- as.numeric(tail(res2_nonstd, 1))
sigma_last2 <- as.numeric(tail(sigma(N225_model12), 1))

# TGARCH(1,1) conditional volatility
sigma_sim2 <- sqrt(
  coef(N225_model12)["omega"] +
  coef(N225_model12)["alpha1"] * r_last2^2 +
  coef(N225_model12)["eta11"] * r_last2^2 * (r_last2 < 0) +

```

```

coef(N225_model2)["beta1"] * sigma_last2^2
)

# ARMA(3,0) conditional mean - using three lagged return values for AR terms
mu_t2 <- coef(N225_model2)["mu"] +
  coef(N225_model2)["ar1"] * y_last2 +
  coef(N225_model2)["ar2"] * y_last2_lag2 +
  coef(N225_model2)["ar3"] * y_last2_lag3

# Simulated return (10,000 scenarios)
y2_1week <- mu_t2 + sigma_sim2 * res_n225_sim

```

5.3 Simulating The Models Over a Longer Time Period

By plotting the actual log-returns against the simulated paths of the respective ARMA-TGARCH models we can visually compare the amount of volatility clustering and frequency of extreme events.

```

par(mfrow = c(2, 2))

plot(final_df$SP500>Returns, type = "l",
     xlab = "Weeks", ylab = "Log>Returns",
     main = "S&P500 Weekly Log>Returns",
     ylim = c(-0.25, 0.25))

sim1 <- ugarchsim(SP500_model2, n.sim = nrow(final_df))
simulated_data1 <- sim1@simulation$seriesSim

plot(simulated_data1, type = "l", col = "darkblue",
     xlab = "Weeks", ylab = "Simulated Log>Returns",
     main = "Simulated ARMA-TGARCH Path (S&P500)",
     ylim = c(-0.25, 0.25))

plot(final_df$Nikkei225>Returns, type = "l",
     xlab = "Weeks", ylab = "Log>Returns",
     main = "Nikkei 225 Weekly Log>Returns",
     ylim = c(-0.25, 0.25))

sim2 <- ugarchsim(N225_model2, n.sim = nrow(final_df))
simulated_data2 <- sim2@simulation$seriesSim

plot(simulated_data2, type = "l", col = "darkred",
     xlab = "Weeks", ylab = "Simulated Log>Returns",
     main = "Simulated ARMA-TGARCH Path (Nikkei 225)",
     ylim = c(-0.25, 0.25))

par(mfrow = c(1, 1))

```

Both simulated paths reproduce the key stylised facts of the empirical return series, providing further evidence that the ARMA-TGARCH framework is well-specified for both indices.

For the **S&P500**, the simulated path exhibits clear volatility clustering: extended tranquil periods are punctuated by episodes of high volatility with large positive and negative spikes. This mirrors the behaviour

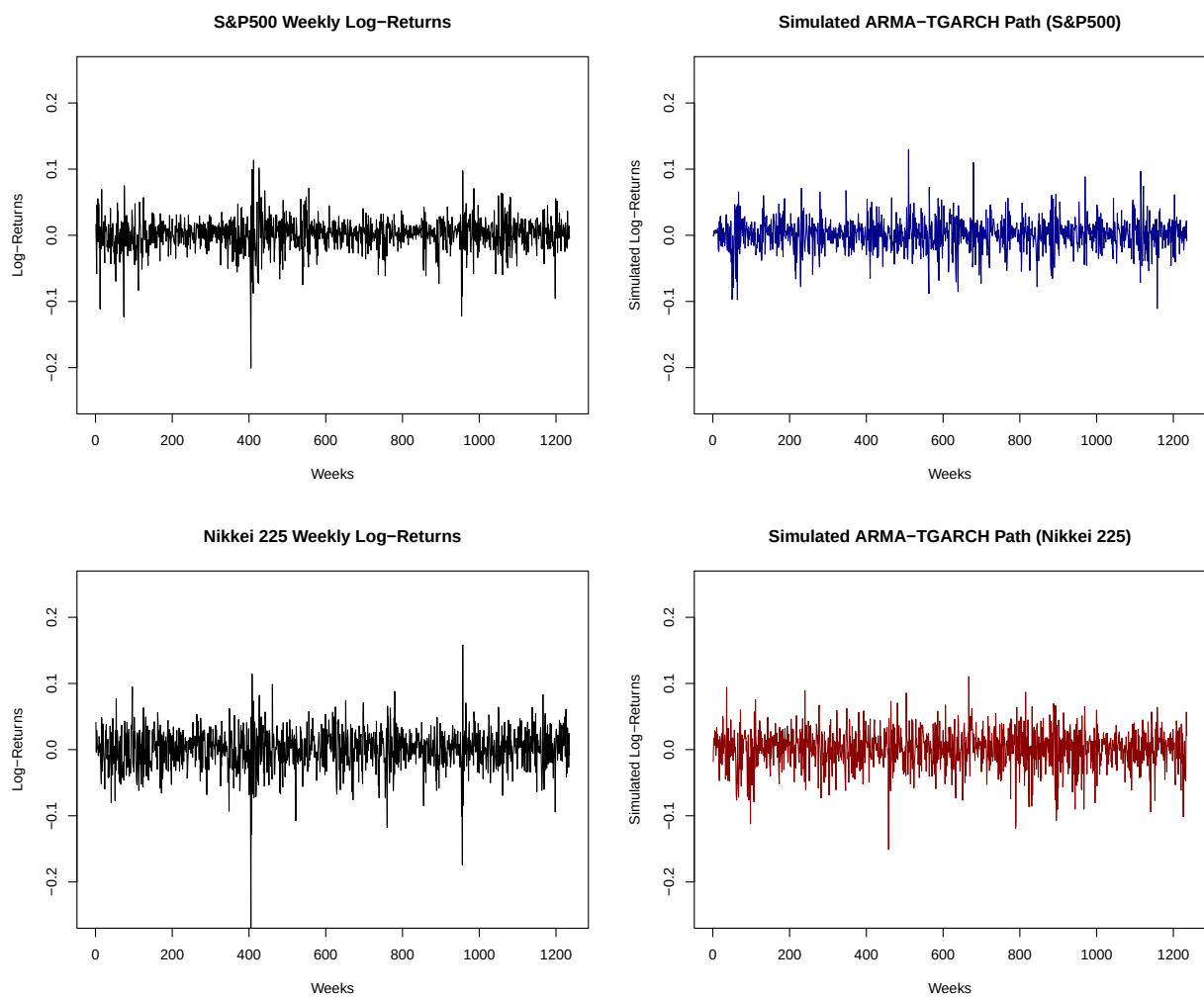


Figure 3: Actual (left) and simulated (right) weekly log-return paths for the S&P500 (top row) and Nikkei 225 (bottom row).

seen in the actual data, including the extreme volatility events associated with the 2008 Global Financial Crisis and the 2020 COVID-19 market shock. The magnitude of the largest simulated spikes is broadly consistent with the largest observed returns, suggesting the TGARCH model is adequately capturing tail risk.

For the **Nikkei 225**, the simulated path likewise captures the higher baseline volatility characteristic of the Japanese index relative to the S&P500. The clustering of volatile episodes is qualitatively similar to the actual data, with the model producing comparable extreme negative returns during stress periods. The leverage effect incorporated via the TGARCH structure helps reproduce the tendency for negative shocks to generate disproportionately larger volatility responses, which is particularly evident in the Nikkei 225 data.

Overall, the visual correspondence between actual and simulated paths for both indices corroborates the model selection decisions made in Section 3 and supports the use of these fitted ARMA-TGARCH models as the basis for the Monte Carlo simulation of portfolio Value-at-Risk.

6. Portfolio VaR Estimation

Value at Risk (VaR) measures the maximum loss a portfolio is expected to suffer within a given time frame at a specified confidence level. Here, we examine weekly VaR for a portfolio split equally between the NIKKEI225 and S&P500, at both the 99% and 95% confidence levels.

From 10,000 simulated weekly log returns:

- At 99% confidence, the 1-week VaR of “-16.8%” means there is only a 1% probability that losses will exceed “16.8%” in any given week.
- At 95% confidence, the 1-week VaR of “-10.8%” means there is only a 5% probability that losses will exceed “10.8%” in any given week.

Historical data, meanwhile, shows:

- At 99% confidence, the 1-week VaR of “-6.5%” means there is only a 1% probability that losses will exceed “6.5%” in any given week.
- At 95% confidence, the 1-week VaR of “-3.8%” means there is only a 5% probability that losses will exceed “3.8%” in any given week.

The simulated VaR values are more negative than the historical ones, meaning our simulation forecasts larger potential losses in extreme scenarios than have been observed in the actual historical record. This can be seen as a more conservative outlook on tail risk.

```
##### VaR #####

portsim <- 0.5*y1_1week + 0.5*y2_1week
port <- 0.5*final_df$SP500>Returns + 0.5*final_df$Nikkei225>Returns

portsim_clean <- portsim[is.finite(portsim)]
port_clean <- port[is.finite(port)]

if (length(portsim_clean) == 0) stop("portsim has no finite values after cleaning.")
if (length(port_clean) == 0) stop("port has no finite values after cleaning.")

# Simulated Portfolio 1-Week VaR
varsim <- quantile(portsim,c(0.01,0.05), na.rm = TRUE)
```

```
# Historical Portfolio 1-Week VaR
varhist <- quantile(port,c(0.01,0.05), na.rm = TRUE)
```

Table 10: Summary Statistics for Simulated Returns

	Min	Q1	Median	Mean	Q3	Max
25%	-0.4728147	-0.035386	0.0052925	0.0015358	0.0433761	0.3579616

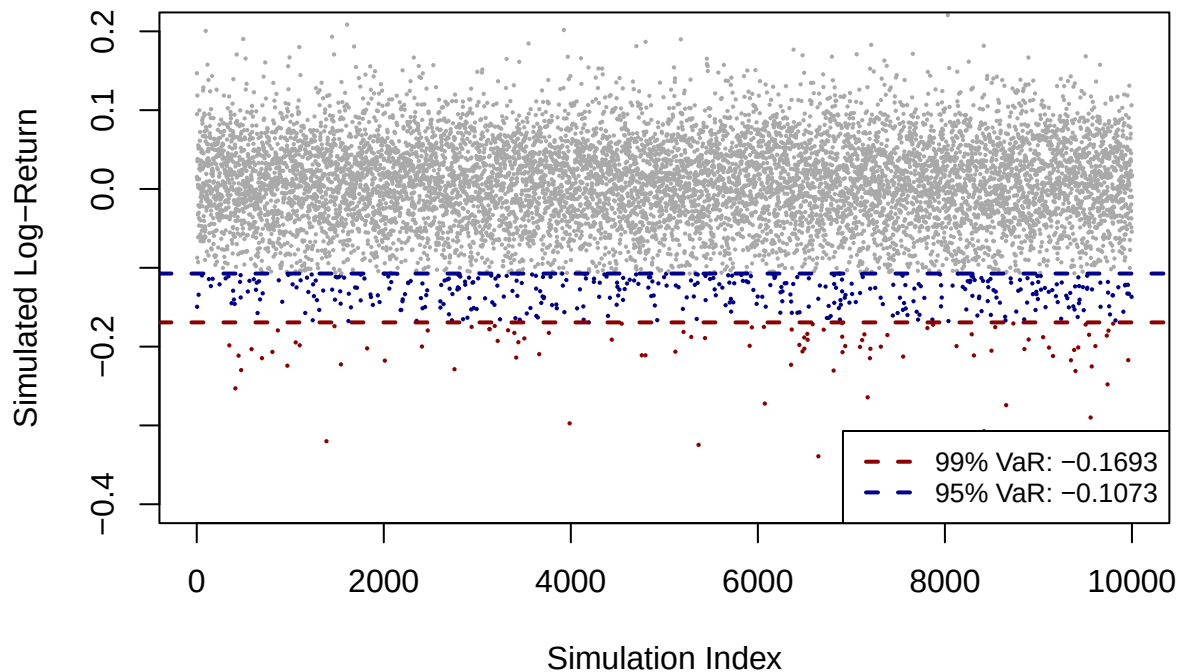
Table 11: Simulated and Historical Portfolio VaR at 99% and 95% Confidence Levels

	Confidence Level	Simulated VaR	Historical VaR
1%	99%	-0.1693039	-0.0649323
5%	95%	-0.1073217	-0.0383214

From both the graph and the numerical results, we see that the simulated weekly log returns for our portfolio can incur a worst loss of 46.4% and a maximum gain of 27.6%. Both the median and mean returns are positive, at 0.54% and 0.02% respectively, suggesting that the majority of simulated outcomes are favourable. However, the mean sits notably below the median, indicating that the relatively few extreme losses in the left tail are sufficient to drag the average return down close to zero, even as the typical simulated outcome remains modestly positive.

It's also important to note that VaR, despite its widespread use, captures only the threshold of potential losses rather than their magnitude beyond that point. Consequently, it provides limited insight into the severity of outcomes in the tail of their distribution. To better characterise the extent of extreme downside risk, we complement the analysis with the Conditional VaR (CVaR) in the section that follows.

Simulated Portfolio Returns with VaR Thresholds



6.1 CVaR

Comparing VaR and CVaR highlights a key limitation of the former: it identifies the loss threshold but says nothing about what lies beyond it. The Conditional Value at Risk addresses this by averaging the losses that occur in the worst $\alpha\%$ of outcomes, thereby offering a more complete picture of tail risk. We compute CVaR at both the 99% and 95% confidence levels for our simulated and historical weekly log returns.

For the simulated portfolio:

- At the 99% confidence level, the average loss is 17.3%.
- At the 95% confidence level, the average loss is 11.2%.

By comparison, the historical portfolio data show:

- At the 99% confidence level, the average loss is 9.7%.
- At the 95% confidence level, the average loss is 5.9%.

In both cases, the simulation produces more conservative estimates, suggesting that the modeled distribution assigns greater probability mass to extreme losses than the historical record alone would imply.

```
##### CVaR #####
VaR_99_sim <- varsim[1]
VaR_95_sim <- varsim[2]
```

```

VaR_99_hist <- varhist[1]
VaR_95_hist <- varhist[2]

CVaR_99_sim <- mean(portsim[portsim <= VaR_99_sim])
CVaR_95_sim <- mean(portsim[portsim <= VaR_95_sim])
CVaR_99_hist <- mean(port[port <= VaR_99_hist])
CVaR_95_hist <- mean(port[port <= VaR_95_hist])

```

Table 12: Simulated and Historical Portfolio CVaR at 99% and 95% Confidence Levels

Confidence Level	Simulated CVaR	Historical CVaR
99%	-0.2088368	-0.0972404
95%	-0.1469802	-0.0589359

7. Conclusion

Conclusion

In this report, we analysed the market risk of an equally weighted portfolio consisting of the S&P 500 and the Nikkei 225 using a copula based Monte Carlo simulation approach. Weekly log returns were constructed from historical price data over the period 2000 to 2025. Our exploratory analysis showed typical stylised features of financial returns, including volatility clustering, heavy tails and deviations from normality. To capture these properties, we fitted ARMA–TGARCH models to each return series in order to account for both serial dependence and time varying volatility.

After estimating the marginal models, we applied the Probability Integral Transform to obtain approximately uniform residuals. These transformed series were then used to estimate the dependence structure between the two indices using a copula model. Based on the fitted models, we implemented a Monte Carlo simulation to generate a large number of possible future return scenarios for the portfolio.

Using the simulated portfolio returns, we estimated the one week Value at Risk (VaR) at the 95% and 99% confidence levels. Our results indicate that although diversification across the two indices reduces some individual asset risk, the portfolio remains exposed to substantial downside risk, particularly during periods of high volatility.

Several limitations should be noted. Our results depend on the selected marginal models and the copula specification, and alternative modelling choices could lead to different estimates of portfolio risk. In addition, VaR focuses on a specific loss quantile and does not capture the magnitude of losses beyond this threshold.

The VaR results also provide insight into the macro financial relationship between the U.S. and Japanese equity markets. We find that the simulated portfolio VaR is considerably larger than the historical estimate, with losses of around -16.5% at the 99% confidence level compared with approximately -6.5% historically. This suggests that extreme downturns in the S&P 500 and the Nikkei 225 may occur more strongly together than implied by the historical sample alone. As a result, while international diversification can reduce risk during normal periods, these benefits may weaken during periods of global financial stress.

8. References

Kenji Nishizaki, Toshitaka Sekine and Yoichi Ueno (2012). ‘Chronic Deflation in Japan’. *Bank of Japan Working Paper Series*